

Answer Key — General Science Grade 10

Sub-Strand 1.3: Nutrition in Animals • Lesson 3: Enzymes and Regions of the Alimentary Canal — The Body's Chemical Toolkit

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	B	C	D	B	A	A

USE AFTER: OBSERVE PHASE — Observe: research groups construct the region-by-region enzyme summary chart (Region | Enzyme Name | Secreted By | Substrate | Product) for the Mouth

Q1. A research group fills in an enzyme chart for the mouth. Which row is correct for salivary amylase?

- A. Secreted by the stomach; acts on protein; produces peptides
- B. Secreted by the salivary glands; acts on starch (for example in ugali); produces maltose**
- C. Secreted by the salivary glands; acts on starch; produces glucose only
- D. Secreted by the pancreas; acts on fats; produces fatty acids and glycerol

Correct answer: B. Salivary amylase is made by the salivary glands and breaks starch into maltose. Glucose is only produced later, after further digestion in the small intestine.

USE AFTER: OBSERVE PHASE — Observe: groups complete the Substrate and Product columns of the enzyme chart for the Small Intestine (pancreatic enzymes)

Q2. Lipase is an enzyme in the small intestine. What does lipase break down, and what does it produce?

- A. Peptides into amino acids
- B. Fats into fatty acids and glycerol**
- C. Proteins into peptides
- D. Starch into maltose

Correct answer: B. Lipase acts on fats (its substrate) and produces fatty acids and glycerol. Maltose comes from amylase, and peptides and amino acids come from protein-digesting enzymes.

USE AFTER: EXPLAIN PHASE — Explain: cold-called group (Group 2 – Stomach) explains what pepsin does to the nyama choma protein

Q3. A learner eats a piece of nyama choma (grilled meat), which is rich in protein. Which statement correctly describes what pepsin does to it?

- A. Pepsin is made in the pancreas and breaks fats into fatty acids and glycerol
- B. Pepsin is made by the gastric glands and breaks protein directly into glucose
- C. Pepsin is made by the gastric glands in the stomach and breaks protein into peptides**
- D. Pepsin is made in the mouth and breaks starch into maltose

Correct answer: C. Pepsin is secreted by the gastric glands, works in the stomach, and its substrate is protein. The product is peptides, not glucose or maltose.

USE AFTER: EXPLAIN PHASE — Explain: teacher's 5-minute consolidating explanation on why enzyme specificity matters (why salivary amylase cannot break down protein)

Q4. Why can salivary amylase in the mouth not break down the protein in nyama choma?

- A. Because salivary amylase is made in amounts that are too small to work on any food
- B. Because chewing destroys salivary amylase before it can act on protein
- C. Because the mouth is too warm for any enzyme to work on protein
- D. Because enzymes are specific: each enzyme acts only on its own substrate, and amylase acts on starch, not protein**

Correct answer: D. Enzyme specificity means each enzyme acts on one type of substrate. Salivary amylase acts on starch, so protein needs a different enzyme such as pepsin or trypsin.

USE AFTER: EXPLAIN PHASE — Explain: cold-called question to Group 3 or 4 on why the small intestine needs so many different enzymes

Q5. The mouth and stomach each have only one or two main enzymes, but the small intestine has many. What is the best explanation?

- A. Every enzyme in the small intestine breaks down all food types, but each one works slowly
- B. The small intestine receives starch, proteins and fats, and each type of food needs its own enzyme**
- C. Enzymes in the mouth and stomach are used up, so the small intestine must replace them
- D. The small intestine is narrow, so it needs more enzymes to push food along

Correct answer: B. Enzymes are specific to their substrates. The small intestine handles carbohydrates, proteins and fats from a mixed meal, so it needs a range of enzymes such as amylase, trypsin and lipase.

USE AFTER: MODEL BUILDING PHASE — Model: adding the enzyme layer and colour-coded key (carbohydrate, protein, fat enzymes) to the cumulative alimentary canal diagram

Q6. When colour-coding an alimentary canal diagram, a learner uses one colour for protein-digesting enzymes. Which pair of enzymes should both be in that colour?

- A. Pepsin and trypsin**
- B. Pepsin and salivary amylase
- C. Pancreatic amylase and pepsin
- D. Lipase and trypsin

Correct answer: A. Both pepsin (stomach) and trypsin (small intestine) act on protein. Amylase acts on starch and lipase acts on fats.

USE AFTER: END OF LESSON — End of lesson: reviewing the completed enzyme chart for the Small Intestine (intestinal enzymes / brush border)

Q7. In the small intestine, intestinal peptidases act on peptides that came from protein digestion. What is the final product of this step?

- A. Amino acids**
- B. Starch
- C. Maltose
- D. Fatty acids and glycerol

Correct answer: A. Intestinal peptidases break peptides into amino acids, the final products of protein digestion that can be absorbed into the blood.